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Taking others into account: combining directly experienced and indirect information in schizophrenia

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An abnormality in inference, resulting in distorted internal models of the world, has been argued to be a common mechanism underlying the heterogeneous psychopathology in schizophrenia. However, findings have been mixed as to wherein the abnormality lies and have typically failed to find convincing relations to symptoms. The limited and inconsistent findings may have been due to methodological limitations of the experimental design, such as conflating other factors (e.g. comprehension) with the inferential process of interest, and a failure to adequately assess and model the key aspects of the inferential process.

Here, we investigated probabilistic inference based on multiple sources of information using a new digital version of the beads task, framed in a social context. Thirty-five patients with schizophrenia or schizoaffective disorder with a wide range of symptoms and 40 matched healthy control subjects performed the task, where they guessed the colour of the next marble drawn from a jar based on a sample from the jar as well as the choices and the expressed confidence of four people, each with their own independent sample (which was hidden from participant view). We relied on theoretically motivated computational models to assess which model best captured the inferential process and investigated whether it could serve as a mechanistic model for both psychotic and negative symptoms.

We found that 'circular inference' best described the inference process, where patients over-weighed and overcounted direct experience and under-weighed information from others. Crucially, overcounting of direct experience was uniquely associated with most psychotic and negative symptoms. In addition, patients with worse social cognitive function had more difficulties using others' confidence to inform their choices. This difficulty was related to worse real-world functioning. The findings could not be easily ascribed to differences in working memory, executive function, intelligence or antipsychotic medication.

These results suggest hallucinations, delusions and negative symptoms could stem from a common underlying abnormality in inference, where directly experienced information is assigned an unreasonable weight and taken into account multiple times. By this, even unreliable first-hand experiences may gain disproportionate significance. The effect could lead to false perceptions (hallucinations), false beliefs (delusions) and deviant social behaviour (e.g. loss of interest in others, bizarre and inappropriate behaviour). This may be particularly problematic for patients with social cognitive deficits, as they may fail to make use of corrective information from others, ultimately leading to worse social functioning.

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Abbreviations: PSP = Personal and Social Performance Scale; TASIT = The Awareness of Social Inference Test

Introduction

Abnormal inference mechanisms in the nervous system resulting in distorted internal models of the world have been argued to be key to schizophrenia.[1] Within this hierarchical Bayesian approach to brain function, beliefs and percepts emerge from an optimal integration of prior expectations and new observations.[2] However, failures in this process may cause abnormal perceptions (hallucinations), abnormal beliefs (delusions) and possibly even negative symptoms such as asociality.[1]

Despite the elegance of this theory of one common mechanism giving rise to such diverse symptoms, empirical evidence for this is yet scarce.[3] Findings have been mixed as to wherein the abnormality lies.[3–10] Furthermore, the identified inferential abnormalities are typically only associated with one type of symptom (e.g. delusions), if any at all.[5,9–11] However, the limited and inconsistent findings could be due to methodological limitations of the experimental design, where other factors such as task comprehension and working memory abilities are conflated with the inferential process of interest, and a failure to adequately model the inferential process.[5,7,9,12–14]

Typically, studies have used variations of the classical beads task to investigate explicit probabilistic inference in schizophrenia. This is a data-gathering paradigm, where participants sample beads from a hidden jar until they feel confident enough to guess which of two jars the beads are sampled from (e.g. the jar with most blue beads or most yellow beads) based on the colour of the sampled beads. The task was originally adapted to assess the relation between inference and delusions, and although patients are consistently found to gather less information and tend to 'jump to conclusions' [i.e. show fewer draws-to-decision (DTD)], the relation between DTD and delusions has been found weak at best.[9,11] In fact, recent studies addressing limitations of the original task have found that both delusion proneness in healthy individuals and delusion severity in patients are associated with increased information-seeking behaviour (i.e. increased DTD).[5,15] Further computational analyses by Baker *et al.*[5] showed that delusional patients displayed stronger reliance on beliefs formed early in the inferential process, or in other words, they had 'sticky' beliefs that were resistant to new evidence.[5,16]

A reliable model of how patients perform sequential integration of one type of information could indeed prove essential in understanding a key abnormality in schizophrenia. However, this research neglects that we usually make inferences based on a multitude of information sources that vary both in type and reliability. This is the case for low-level perception,[17] but perhaps even more so for our social lives, where we constantly have to deal with different people that say or do different things with different degrees of certainty. This information has to be combined with our own experiences and beliefs, which may be more or less certain. In short, the classical beads task fails to address how we simultaneously balance and integrate heterogeneous information sources. This is of particular relevance in schizophrenia, where patients typically struggle with such everyday situations. Schizophrenia has even been described as a disorder of self-other processing, displaying problems with self-other integration and distinction that range from self-disturbances (e.g. delusions of being controlled) to altered mimicry (e.g. echolalia) and difficulties inferring others' mental states and distinguishing them from one's own (i.e. mentalizing deficits).[18]

Indications of inferential problems with multiple sources were reported in a recent study by Jardri *et al.*[3] In this study, participants had to integrate two different sources of information, which varied in degree of certainty, in order to decide which lake a red fish was from. The two sources of information were: a prior (the basket size associated with each of two lakes, reflecting the probability of catching a red fish in them) and sensory evidence (the distribution of red and black fish in each lake). Interestingly, patients relied more on the most recent evidence (contrary to the findings of Baker *et al.*[5]) and this tendency was related to psychotic symptoms. There may be several reasons for this. First, participants could only see the fish distributions in the lakes (sensory evidence) and not the basket sizes (prior information) when making the decision. Patients may have disregarded the prior information due to working memory deficits, which are prevalent in schizophrenia.[19] Indeed, weighing of prior information was associated with working memory performance.[3] Second, the task is highly abstract. It may not be clear to patients how they should interpret and use the basket size compared to the number of red fish in the lakes. Such comprehension difficulties may contribute to larger reliance on sensory evidence. Finally, the task design conflates information type (basket or lake) and time (prior or new evidence). Thus, it is possible that if the lakes and baskets were presented in reversed order or simultaneously, we would still see a larger reliance on the lake distributions, i.e. it may be qualities of the information source rather than order that matters.

To address these alternative interpretations and directly investigate probabilistic inference based on multiple sources of information, we used a new version of the beads task.[21] On each trial, participants had to guess the colour (red or green) of the next marble drawn from a hidden jar based on a sample from the jar (eight marbles) and the choice and confidence of four other people. Each of the others' choices were said to be based on their own sample of eight marbles from the same jar. By systematically varying the evidence for a specific colour in the sample and the others' choices, we could assess how patients integrate different sources of information varying in degree of certainty. Importantly, all information was present when they had to make a choice to avoid confounding inference with working memory function. To ease task comprehension and make the task less abstract, we gave careful task instructions (see 'Materials and methods' section) and framed the task in a social context, since reasoning is known to be easier in social contexts,[22] thus reducing such potential confound of misunderstanding. Altered inference may become particularly evident in social contexts,[23] given the pronounced social cognitive deficits in schizophrenia[24,25] and the very social nature of many symptoms (e.g. hearing voices, feeling persecuted, asociality).[26]

To investigate the patients' information integration processes, we relied on theoretically motivated computational models based on Bayes theorem. A common concern in the study of mental disorders is the lack of adequate modelling of behavioural data, with strong arguments in favour of theoretically motivated computational models replacing more traditional statistical techniques.[27] In this study, we compared four different Bayesian models as well as two control models: (i) simple Bayes; (ii) weighted Bayes; (iii) circular inference with; and (iv) without interference; (v) a heuristic model; and (vi) the generalized linear model (GLM) (for details, see 'Materials and methods' section and Supplementary material). The Bayesian framework provides a formal solution to optimal integration of information under uncertainty where each source of information is weighed according to its precision or reliability. While simple and weighted Bayes are established models to integrate information, circular inference is a biologically informed model that is shown to account for psychotic symptoms.[3,20] The model builds on the theory that a failure to maintain the balance in excitatory/inhibitory neural processing may cause reverberation of sensory and prior information, so that sensory information is misinterpreted as prior beliefs (or vice versa). The net result is an overcounting of potentially unreliable information, which might lead to abnormal perception (hallucinations) and abnormal beliefs (delusions).[3,20]

We compared patients' performance to a matched healthy control group, and we investigated whether symptoms and social functioning were related to how patients integrated information. In addition, we assessed whether altered inference was related to social cognitive deficits, a relationship that is not yet well understood.[6,28–30] Specifically, we assessed whether there was a relation between inferences made in the two contexts, i.e. inferences based on integration of direct experience and information from others, and mental state inferences based on integration of multiple social cues. Finally, we assessed the role of potential confounders (working memory, executive function, intelligence, antipsychotic medication). We hypothesized that patients would discount information from others in favour of directly experienced information (i.e. their own sample) relative to controls, and that this information integration style would be related to more severe psychotic symptoms, more severe negative symptoms, larger social cognitive deficits and lower social functioning.

Materials and methods

Participants

Forty patients with an ICD-10 DCR diagnosis of schizophrenia or schizoaffective disorder and 40 matched healthy control subjects were included in the study (recruitment period: June 2013 to August 2014). Diagnosis was confirmed using the Schedules for Clinical Assessment in Neuropsychiatry (SCAN).[31,32] Controls were pairwise matched to the patients according to age, gender, childhood residence as well as commenced educational level (or parents' educational attainment if higher than the patient's) and parental socioeconomic status when possible (Table 1).

Patients were recruited through the Psychiatric Centre of the National Hospital of the Faroe Islands. Control subjects were contacted based on age and gender and, if they fulfilled the inclusion criteria and matched a patient, were offered to participate in the study. Participants were between 18 and 55 years old. The inclusion criteria were: no current psychoactive substance use disorders (except nicotine), no head trauma, neurological or medical disorder that could affect brain functioning, and an estimated IQ >70 based on prior history or testing. In addition, the controls should not be taking psychotropic drugs or have a history of severe mental disorder either among themselves or a first-degree relative. The participants were screened for recent use of psychoactive substances (THC/cannabis, opiates, amphetamine, MDMA, benzodiazepines, cocaine) using a urine stick (NanoSticka® 200-32). Patients without a prescription who had a positive test were excluded. None of the controls had a positive test.

Based on pilot testing and a previous study using a similar paradigm,[21] we aimed to include 40 participants in each group to allow up to 20% drop-out. A total of 56 patients and 45 controls were invited to the initial interview. Of these nine declined to participate (six patients, three control subjects), seven participants (five patients, two control subjects) did not fulfil the inclusion criteria and were therefore excluded, and five patients dropped out during initial interviewing. Two additional patients were excluded after initial inclusion as it became clear that they did not fulfil the inclusion criteria. In addition, two patients did not perform the task because they did not understand the task instructions and one patient opted out due to lack of time. Thus, data from 35 patients (four with schizoaffective disorder) and 40 controls were included in the analyses. Thirty-two of these patients were taking antipsychotic medication at the time of testing. Antipsychotic dose was converted to chlorpromazine equivalents.[33,34]

General procedure

The task was administered as part of a larger battery of cognitive tests. In addition, symptom severity and level of functioning were assessed with the Scale for the Assessment of Positive/Negative Symptoms (SAPS/SANS)[35,36] and the Personal and Social Performance Scale (PSP),[37] respectively. We used the Danish version of 'The Awareness of Social Inference Test' (TASIT, Part 2: Social Inference–Minimal)[38] to assess social cognition. It consists of short video vignettes depicting everyday interactions and participants answer yes/no questions about the protagonists' thoughts, feelings, intentions and meaning.[39,40] The test requires social cue integration (e.g. intonation, facial expression, body language, verbal content) in order to infer when the protagonists are being sincere or sarcastic. We used the TASIT 2A total score.

We used four subtests (Vocabulary, Similarities, Block Design, and Matrix Reasoning) of the Wechsler Adult Intelligence Scale-IV[41] to estimate IQ. Working memory and executive function were assessed with Spatial Working Memory and the Intra-Extra Dimensional Set Shift (IED) from the Cambridge Neuropsychological Test Automated Battery (CANTAB),[42] respectively (Table 1). The study was performed in accordance with the relevant national guidelines and regulations and with the Declaration of Helsinki. Written informed consent was obtained from all participants after the procedure had been explained.

Table 1 Characteristics of the participants

	Schizophrenia Mean (SD)	Controls Mean (SD)
Demographics		
Sample size, *n*	35	40
Age	38.0 (10.7)	39.3 (10.5)
Sex, males/females, *n*	22/13	27/13
Handedness, right/left, *n*	32/3	36/4
Educational level commenced[a]	2.1 (0.7)	2.3 (0.7)
Years of education	12.6 (2.7)	14.3 (3.1)
Parental socioeconomic status[b], high/middle, *n*	12/23	13/27
Cognitive functions		
Estimated IQ	93.3 (14.4)	102.5 (13.8)
The awareness of social inference test, total score	44.5 (7.3)	49.7 (5.4)
Spatial Working Memory, total errors	16.1 (10.5)	7.6 (9.0)
Intra Extra Dimensional Set Shift, total errors adjusted	42.1 (39.3)	18.6 (16.1)
Level of functioning		
Personal and Social Performance Scale	59.8 (13.7)	86.1 (5.1)
Living independently/with parents/in institution, *n*	13/15/7	37/3/0
Early retirement or other financial support[c], *n*	30	0
Illness duration, psychopathology, medication status		
Illness duration in years	14.7 (9.8)	–
SAPS total score [range]	4.20 (4.13) [0–13]	–
Global rating of severity of hallucinations [range]	1.31 (1.94) [0–5]	–
Global rating of severity of delusions [range]	1.89 (2.03) [0–5]	–
Global rating of severity of bizarre behaviour [range]	0.49 (0.92) [0–3]	–
Global rating of positive formal thought disorder [range]	0.51 (0.98) [0–3]	–
SANS total score [range]	7.66 (4.63) [0–20]	–
Global rating of affective flattening [range]	1.34 (1.11) [0–4]	–
Global rating of alogia [range]	0.86 (1.31) [0–4]	–
Global rating of avolition—apathy [range]	2.34 (1.21) [0–5]	–
Global rating of anhedonia—asociality [range]	1.97 (1.48) [0–5]	–
Global rating of attention [range]	1.14 (1.40) [0–4]	–
Chlorpromazine equivalent dose	703 (577)	–

SANS = Scale for the Assessment of Negative Symptoms; SAPS = Scale for the Assessment of Positive Symptoms; SD = standard deviation.
[a]Commenced educational level divided into four levels: 1: primary school (up to 10 years of education), 2: secondary school/professional training, 3: bachelor programme or shorter further education, 4: master programme.
[b]Parental socioeconomic status (SES) was based on the parent with the highest education and an estimated salary according to their employment. It was divided into three levels: low, middle, high. None of the parents had a low SES.
[c]For example, cash benefits, sickness benefits.

Participants received a fixed amount in compensation (gift card) for participating in the study.

Experimental paradigm

We used a shortened version of the urn task described in Campbell-Meiklejohn *et al.*[21] To ensure task comprehension, participants received detailed task instructions, practice trials and debriefing after the task. Although the jars were always hidden behind cardboard during the task (Fig. 1), during task instructions, participants were shown a picture of many jars containing different distributions of red and green marbles, to make it clear that all jars were full of marbles and contained different distributions. In addition, they performed an individual pre-task before the actual task. It contained 16 trials plus five practice trials. On each trial, they were presented with a random sample of eight marbles (zero to eight red and green marbles) from a hidden jar and were instructed to infer the colour (red or green) of the next marble drawn from that specific jar as well as state their confidence in that choice (high or low). The marbles were said to be replaced before the decision had to be made. This pre-task was used to make sure that the participants understood the responses of the other people (agents) during the following task, including how they might be using confidence.

During the experimental task, participants were again presented with a new hidden jar on every trial and were instructed to infer the colour of the next marble drawn from that jar. Before deciding, however, participants were presented with a sample of eight marbles and the decisions of four other people (agents), which each stated their choice as well as the confidence in that choice (Fig. 1). Others' decisions were said to be based on their own sample of eight marbles randomly drawn from the same jar and all the marbles were said to be replaced before the participant had to make a decision. Due to the specific patient group, we deliberately avoided providing an elaborate cover story about these other people. Agents' confidence was communicated as an explicit cue (identical to the response options during the pre-task): the smug (confident cue) or perplexed (unconfident) animated smiley that appeared within the indicated choice (red or green marble) (Fig. 1). To avoid that participants made hasty guessing to reduce task duration, all information was available on the screen for 500 ms before a decision could be made. After a choice, confirmation of the choice was displayed. No feedback was provided. The task consisted of 105 trials (plus 10 practice trials) that were generated by creating every possible combination, across trials, of: the number

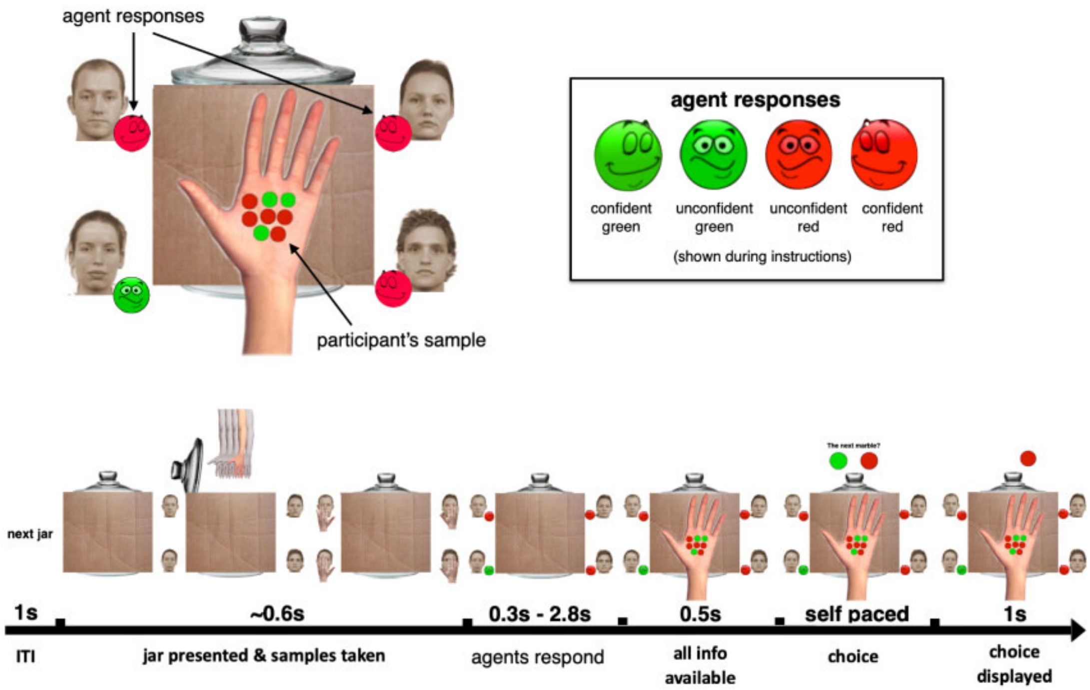

Figure 1 The urn task. Modified from Campbell-Meiklejohn *et al.*[21] At the beginning of each trial a new jar was presented following the words 'next jar'. This text was presented for 1000 ms and served as a reminder to the participants that each jar was different. During the 'pre-task', the jars were numbered, to make this explicit. When a new jar appeared, the contents were hidden. A display of five hands reaching for five different samples from the jar was then animated. Next, the decisions of the four agents were shown with their confidence in those decisions. Agent faces were represented with neutral expression. To avoid association between answers and specific agents, agents were randomly drawn from a set of 30 faces on each trial. Agent answers were expressed by the colour of the circle next to their faces. Confidence was indicated as the expression within the circle as well as the speed at which the answer was shown (rapid + smug = high confidence). Individual agent answers took between 300 and 2800 ms to appear. Next, the participant's own sample of marbles appeared on the screen and was displayed with all other information for 500 ms before the participant could make his/her choice (red or green). The chosen colour was highlighted for 1000 ms, followed by the next trial.

of others choosing red or green, confidence of others choosing red or green, and participants' samples—all varying independently of one another. For each number of others choosing red (0–4), there were 21 trials—three for every possible participant sample (1–7 reds). To reduce the number of necessary trials, agents that chose the same colour had the same confidence level. The trials were randomized. The task had no extrinsic gain or loss. There was no explicit or implicit cost to weighting one's sample and the information from the agents, other than the intrinsic incentive to provide a correct response.

Statistical analyses and computational modelling

We implemented four theory-driven models of information integration,[3] as well as two control models. The models are briefly described here and the full implementation is detailed in the Supplementary material. The simple Bayes model assumes that the directly experienced information (i.e. the participant's sample) and the information from the agents are fully trusted, thus weighing them equally, i.e. one's own sample weighs as much as the information provided by the choice of one of the agents. In this situation, the optimal form of inference is to directly combine the sources of information using Bayes' theorem. Note that the simple Bayes model—although it is unlikely to capture the actual decision-making process—is the best model when participants treat the different sources of information equally, and the model acts as a stepping stone to understand and build the more complex models. The weighted Bayes model builds on the previous model, but allows the participant to attribute different degrees of trust to the directly experienced information and the agents' information and therefore to weigh their impact on the decision differently. Note that when the two weights are equal and fully trusted (w_{Self} = w_{Other} = 1), the weighted Bayes model is equivalent to the simple Bayes model (Supplementary Fig. 1). The circular inference model builds on the weighted Bayes model by adding two components: (i) a free parameter per information source, indicating the strength of the overcounting (alpha); and (ii) interference between the sources of information. In the weighted Bayes model, directly experienced information and information from the agents are considered only once, while in the circular inference model, the information can be counted multiple times. Note that even if a source of information might be considered multiple times, participants do not need to trust it fully, thus maintaining meaningful weight parameters. We also tested whether the model without the interference term could describe the data just as well, to avoid unnecessary complexity (Supplementary Fig. 8). The circular inference model is equivalent to the weighted Bayes model when both alphas (α_{Self} and α_{Other}) are equal to one and no interference is present. If in addition, the weights are both equal to one, the model is equivalent to the simple Bayes model (Supplementary Fig. 2). In order to make our findings more easily comparable with more traditional statistical approaches,[21] we also included a logistic regression model. Finally, previous studies suggest that participants may use simpler reasoning strategies,[7,14,43] we therefore assessed whether participants were using simple heuristics to solve the task as opposed to the Bayesian models presented above. The heuristic model had the following probabilistic decision rules and always relied on one type of information only (i.e. no information integration): (i) choose the colour that is most of in the sample, if indecisive; then (ii) choose the colour that most others choose, if indecisive; then (iii) choose the preferred colour (bias).

We used R, Stan and brms[44,45] to implement the models. All models were implemented in a Bayesian multilevel fashion: partially pooling information from all participants to improve our inferences and explicitly accounting for the uncertainty in the parameter estimates. For comparison we also report individually fitted models (no pooling of information) in the Supplementary material. We chose to use regularizing priors, that is, priors weakly sceptical of the effects, to decrease chances of model overfitting.[46] We compared the models using estimated out-of-sample error (Pareto-smoothed Information Sampling Leave-One-Out Information Criteria, PSIS-LOO),[47] which estimates the likely generalizability of the model to new data. We also performed tests of model and parameter recovery, reported in Supplementary Tables 1 and 2 and Supplementary Figs 3–5.

Cognitive and clinical features in parameter estimation

After identifying the best fitting model, we further assessed in this model whether cognitive and clinical factors were related to the patient's use of information. In particular, we assessed whether cognitive function (e.g. TASIT, IQ), specific symptom clusters (e.g. hallucinations, delusions), general symptom severity (SANS or SAPS total score), antipsychotic medication dose (chlorpromazine) and level of functioning (PSP) would modulate the integration of the directly experienced and the indirect information, by affecting the model parameters. All of these analyses were conducted on the patients only (and, when relevant, in controls only for baseline comparison) by conditioning the estimated parameters on each patient and the additional variable of interest (i.e. we expected the parameters to vary by individual—random effect—and by, for example, global rating of hallucinations—fixed effect). For ease of comparison with previous studies not relying on multilevel modelling, and in order to assess the contribution of a symptom cluster when adjusting for the others, we also report the analysis of the relation between symptoms and individual-level fitted model parameters in the Supplementary material.

Evidence ratio

To quantify the support for our hypotheses, we calculated an evidence ratio in the form of the posterior probability of the directed hypothesis (e.g. parameter > 0) against the posterior probability of the opposite hypothesis (e.g. parameter $\leqslant 0$)[48] with the following interpretation of evidence: 1–3 = anecdotal; 3–10 = moderate; 10–30 = strong.[49] Additionally, we report a credibility score for the hypothesis, that is, the proportion of the samples from the posterior conforming to the prediction.

Data availability

The dataset analysed is available from the corresponding author on reasonable request. The implementation code is available at https://github.com/fusaroli/TakingOthersIntoAccount.

Results

Model comparison

We first compared the six models (simple Bayes, weighted Bayes, circular inference with and without interference, GLM and heuristic). The circular inference model with interference was the most likely given the data across groups, credibly minimizing estimated out-of-sample error. PSIS-LOO indicated credible differences from all other models. This was also the case within each group (Table 2). Individual level fitted models confirmed that circular inference (with or without interference) was most likely for 70 of 75 participants (Supplementary material). We therefore focused on the circular inference model, but parameter estimates for all models are available in Supplementary Tables 3–8. Note that model recovery tests indicate that were the true model a circular inference without interference, our model comparison procedure would not perfectly differentiate between circular inference with and without interference (60% accuracy, see details in Supplementary material). Therefore, further interpretation of which participants are identified as best described by the model with or without interference is not warranted.

Schizophrenia and circular inference

Participants generally took confidence into account, did not take any source of information at face value (weights <1, indicating uncertainty in the use of information), and overcounted information (loop parameters >1, indicating that participants paid more attention to the direction of the evidence than to small differences in it). Participants were also more affected by directly experienced than by information from others (indirect information), but patients more so than control subjects. Specifically, patients over-weighed and overcounted directly experienced information [mean weight: 0.92, 95% credible interval (CI) 0.90 0.93, in patients versus 0.89,

Table 2 Model comparison

Model	All participants			Schizophrenia			Controls		
	ΔELPD	SE	Weight	ΔELPD	SE	Weight	ΔELPD	SE	Weight
Simple Bayes	−3581.4	125.9	0	−2639.9	108.4	0	−968.5	60.6	0
Heuristic model	−2052.5	46.1	0	−1482.2	65.1	0	−573.5	40.1	0
Weighted Bayes	−466.2	23.0	0	−349.7	18.0	0	−146.2	13.8	0
GLM	−56.0	18.6	0.33	−92.9	12.3	0	−45.6	13.1	0.13
Circular inference no interference	−81.9	13.5	0	−7.1	8.7	0.43	−55.1	6.6	0
Circular inference	**0**	**NA**	**0.67**	**0**	**NA**	**0.57**	**0**	**NA**	**0.87**

ELPD = the expected log predictive density of the model, calculated using PSIS-LOO. In other words, it indicates the estimated out-of-sample error of the model: the relative likelihood of the model given the data, adjusting it for potential overfitting and influential data-points. ΔELPD indicates the difference in ELPD compared to the best model: the lower the score, the bigger the distance from the best model. SE = standard error in the estimate of the difference. Weight indicates the stacking weights based on PSIS-LOO, that is, how strong the preference for a given model should be. 0 indicates a very unlikely model, 1 the most likely model having no likely competitors amongst the models analysed, intermediate values indicate relative uncertainty as to which model is best.[50] GLM = generalized linear model; NA = not applicable.

95% CI 0.87 0.90, in controls; mean loops of 6.24, 95% CI 5.13 8.21, in patients versus 2.66, 95% CI 2.15 3.23, in controls]. In other words, small amounts of evidence in their samples (e.g. six red marbles instead of five) made a larger difference in the patients' propensity to guess that the next marble would be red, than it did for control subjects. Patients also underweighted indirect information (both choice and confidence of others) more than controls, mean weight for others of 0.64 (95% CI 0.62 0.66) for high confidence and 0.57 (95% CI 0.54 0.59) for low confidence in patients versus 0.74 (95% CI 0.72 0.76) and 0.62 (95% CI 0.60 0.64) for high and low confidence in control subjects, but showed similar overcounting (2.05, 95% CI 1.41 3.18, versus 2.01, 95% CI 0.99 4.21) (Table 3 and Fig. 2).

Cognitive function and circular inference

There was strong evidence for an association between TASIT performance and the use of others' confidence in informing choice behaviour (Supplementary Table 12, evidence ratio > 1000, credibility = 1), i.e. the better patients were at inferring others' mental states, the more they relied on others' confidence to weight information from them. In particular, patients with the minimum TASIT score would weight low and high confident others 0.55 and 0.58, respectively; while patients with the maximum TASIT score 0.58 and 0.68, respectively.

We assessed whether this could be related to IQ, since IQ is known to be related to TASIT performance.[40] Indeed, there was a similar association, i.e. the higher the patients' IQ, the more they relied on others' confidence to weight information from them (Supplementary Table 14, evidence ratio > 1000, credibility = 1). In particular, patients with the minimum IQ score would weight low and high confident others 0.54 and 0.57, respectively; while patients with the maximum IQ score 0.58 and 0.69. However, when adjusting for IQ, there was still strong evidence for a positive association between TASIT performance and use of others' confidence (Supplementary Table 16, β = 1.02, 95% CIs = 0.52 1.52, evidence ratio > 1000, credibility = 1). Interestingly, we saw the same pattern in the control subjects, i.e. both TASIT and IQ were uniquely related to weighing of others' confidence (Supplementary Tables 13, 15 and 17). In particular, control subjects with the minimum TASIT score would weight low and high confident others 0.59 and 0.64, respectively; while those with the maximum TASIT score 0.64 and 0.78 (Supplementary Table 13, evidence ratio > 1000, credibility = 1). Control subjects with the minimum IQ score would weight low and high confident others 0.61 and 0.69, respectively; while controls with the maximum IQ score 0.63 and 0.75 (Supplementary Table 15, evidence ratio > 1000, credibility = 1).

There was no evidence for a negative association between loops for self and IQ (evidence ratio = 0.08, credibility = 0.08). In fact, the evidence was in the opposite direction, i.e. the higher the patient IQ, the higher the loops for self (Supplementary Table 14, mean loops for self changing from 3.78 to 5.05 from lowest to highest IQ, evidence ratio = 12.3, credibility = 0.92). In contrast, there was no evidence for an association between loops for self and IQ in the control subjects (Supplementary Table 15). There was no credible evidence that working memory or executive function were associated with patients' task performance (Supplementary Tables 18 and 19). Model comparison also showed that models including measures of IQ, spatial working memory or executive function (IED) gained 0 weight, while the model including TASIT was the most likely given the data (Supplementary Table 20, stacking weight TASIT model: 0.89; baseline model: 0.11).

Psychopathology and circular inference

There was strong evidence for a positive association between loops for self and hallucinations (5.42 with a score of 0, 7.00 with a score of 5, evidence ratio = 18.1, credibility = 0.95, Supplementary Table 21), delusions (5.31 to 6.62, evidence ratio = 11.7, credibility = 0.92, Supplementary Table 22), bizarre behaviour (5.36 to 7.02, evidence ratio = 20.6, credibility = 0.95, Supplementary Table 23) and SAPS total score (5.21 to 7.1, evidence ratio = 29.9, credibility = 0.97, Supplementary Table 25), while it was moderate for formal thought disorder (5.58 to 6.69, evidence ratio = 5.5, credibility = 0.85, Supplementary Table 24). Similarly, there was strong evidence for a positive association between loops for self and anhedonia-asociality (5.21 to 7.1, evidence ratio = 28.7, credibility = 0.97, Supplementary Table 29), avolition-apathy (5.31 to 6.89, evidence ratio = 12.0, credibility = 0.92, Supplementary Table 28), attention (5.42 to 6.75, evidence ratio = 10.5, credibility = 0.91, Supplementary Table 30) and SANS total score (5.26 to 6.82, evidence ratio = 13.7, credibility = 0.93, Supplementary Table 31), while it was moderate for affective flattening (5.47 to 6.36, evidence ratio = 4.2, credibility = 0.81, Supplementary Table 26) and not credible for alogia (5.37 to 5.70, evidence ratio = 1.6, credibility = 0.62, Supplementary Table 27). For the other model parameters, there was either no or only moderate evidence for associations with different symptoms (Supplementary material).

Table 3 Parameter estimates for the circular inference model including group

Model	Mean	2.5% CI	97.5% CI	Evidence ratio
Bias for red (controls)	−0.07	−0.16	0.02	
Group-level diff in bias for red	0.01	−0.10	0.13	
Confidence (controls)	2.07	1.85	2.30	
Group level diff in confidence	**−0.18**	**−0.48**	**0.13**	**7.1, 88% credibility**
Weight for self (controls)	1.25	1.06	1.44	
Group level diff in weight for self	**0.39**	**0.11**	**0.67**	**362.6, 100% credibility**
Weight for others (controls)	−2.19	−2.41	−1.98	
Group level diff in weight for others	**−0.61**	**−0.91**	**−0.32**	**> 1000, 100% credibility**
Loops for self (controls)	0.98	0.77	1.17	
Group level diff in loops for self	**0.88**	**0.61**	**1.17**	**> 1000, 100% credibility**
Loops for others (controls)	0.72	0.21	1.16	
Group level diff in loops for others	−0.02	−0.60	0.58	1.1, 53% credibility

Note that the parameters can be on transformed scales, e.g. weights are on a log odds scale, and loops on an exponential scale. See the 'Materials and methods' section and the Supplementary material for more details.

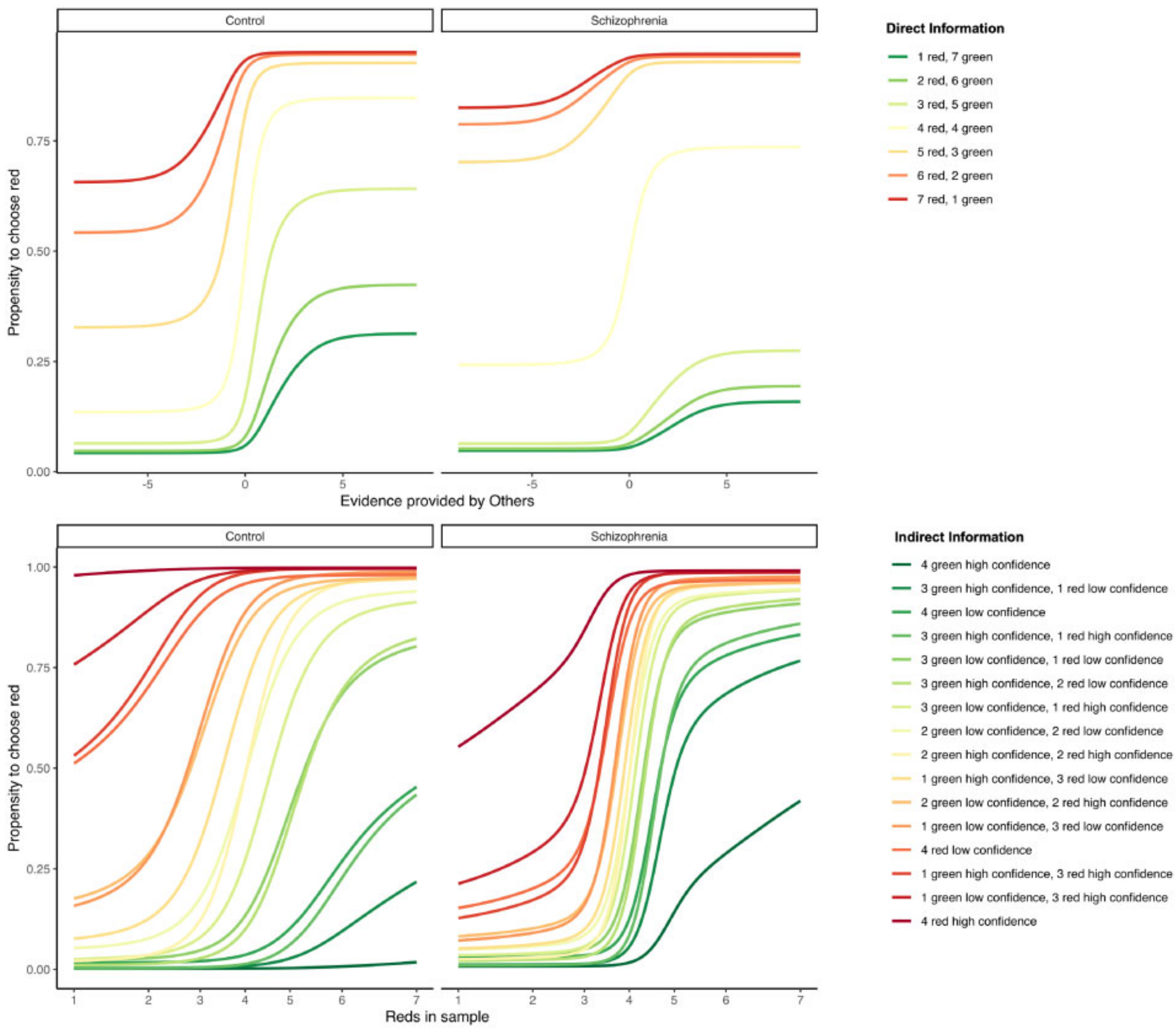

Figure 2 Information integration in patients with schizophrenia and matched healthy controls. The *top* plots represent the influence of the agents on controls (*top left*) and patients (*top right*). The *x*-axis indicates the amount of combined evidence for red provided by the agents on a log-odds scale, the *y*-axis indicates the probability of choosing red. The colour indicates the level of evidence for red in the sample. The average patient showed less influence from others (shallower slopes) and more influence from directly experienced information (more spread curves). Note that for simplicity of representation we collapsed others' confidence and choices, to construct a continuous one-dimensional scale of indirect evidence for red. The *bottom* plots represent the influence of directly experienced information on control subjects (*bottom left*) and patients (*bottom right*). The *x*-axis indicates the number of red marbles contained in one's own sample, the *y*-axis indicates the probability of choosing red. The colour indicates the level of evidence for red provided by the other agents. Note how the coloured lines are steeper and more collected in the average patient compared to the average control subject: this indicates a more decided role of directly experienced information (steeper slopes) and a lower influence of indirect information (as the lines are more similar to each other, no matter the level of indirect evidence). See Supplementary Figs 6, 7A and B for participant-level representations of the model.

We assessed whether medication could be confounding the results. The evidence for a positive association between chlorpromazine and loops for self was only moderate (evidence ratio = 3.1, credibility = 0.75, mean loops changing from 5.5 for no medication to 6.3 for highest medication dose, Supplementary Table 32) and could just as well reflect disorder severity, where the evidence was much stronger.

Finally, we also assessed parameters fitted at the individual level (no pooling), as done in previous studies.[3] Here, there was in most cases strong evidence for a positive association between loops or weight for self and all symptom clusters (except for formal thought disorder). There was a similar picture for loops and weight for others, where there was moderate to strong evidence for a negative association with all symptoms (except formal thought disorder). On the other hand, there was no credible evidence for an association between confidence and symptoms, expect for bizarre behaviour, where the evidence was moderate (Supplementary Table 37). Importantly, since some of the symptoms covaried (see correlation matrix S39 and network structure in Supplementary Fig. 9), we also assessed associations between the model parameters and each psychotic or negative symptom cluster while controlling for the other psychotic or negative symptom clusters, respectively. Many of the associations survived this correction. For instance, hallucinations, delusions, bizarre behaviour, anhedonia-asociality, alogia and attention were uniquely associated with loops for self. Similarly, hallucinations, delusions, affective flattening, avolition-apathy and attention were uniquely associated with weight for others (for full details see Supplementary Table 38).

Level of functioning and circular inference

There was a similar pattern for PSP and IQ. Specifically, the higher the level of functioning, the more patients relied on confidence. Patients with minimum PSP score used an average weight of 0.56 for low confident and 0.61 for high confident others; while patients with the highest score 0.57 and 0.64, respectively (Supplementary Table 33, evidence ratio = 36.6, credibility = 0.97). This was the case even when controlling for IQ (evidence ratio = 4.37, credibility = 0.81, Supplementary Table 35). In addition, there was moderate evidence for a positive association between PSP and loops for self (evidence ratio = 8.7, credibility = 0.90, Supplementary Table 33), which disappeared when controlling for IQ (evidence ratio = 1.15, credibility = 0.54, Supplementary Table 35). For control subjects, we saw a similar association between PSP score and confidence, also when adjusting for IQ, but not between PSP and loops for self (Supplementary Tables 34 and 36). Controls with minimum PSP score used an average weight of 0.61 for low confident and 0.69 for high confident others; while controls with the highest score 0.63 and 0.74, respectively (evidence ratio > 1000, credibility = 1, Supplementary Table 34).

Control experiment

We ran a control experiment in healthy individuals (Supplementary material) to assess whether framing the indirect source of information in a social or non-social context impacted task performance. Results showed that while there was no credible difference in how they used their own sample in the two contexts, they did rely more on a confident person than on a confident algorithm (Supplementary Table 40).

Discussion

We investigated the hypothesis that the diverse symptoms characteristic of schizophrenia could stem from a common abnormality in inference mechanisms.[1] This was done using a new version of the beads task, designed to investigate inference based on concurrent integration of multiple sources of information. By comparing different computational models of how inference could be altered in schizophrenia, we found that different degrees of circular inference best described choice behaviour in both patients and controls. This is concordant with circular inference being the best model to describe the integration of sensory and prior information in a previous study and suggests that overcounting of information is not pathological *per se* and may even be a part of normal neural functioning.[3,51]

Patients tended to put more weight on and overcount directly experienced information compared to control subjects and, conversely, they tended to put less weight on the information from others (indirect information). The findings are in line with the study by Jardri *et al.*,[3] which found that patients put more weight on and overcounted sensory evidence and less weight on prior information. Importantly, our results suggest that overcounting of direct experience may not only be a key mechanism underlying psychotic symptoms, as previously reported,[3] but also negative symptoms. The fact that overcounting of direct experience was associated with most symptom clusters cannot be simply ascribed to general disorder severity or high correlations between symptoms. Although some symptoms were correlated, many of the psychotic and negative symptoms were uniquely related to overcounting of direct experience. Over-weighing of direct experience as well as under-weighing and under-counting of indirect information could also play a role in the psychopathology, although the evidence for this was not as strong.

We framed the task in a social context to make it less abstract, but also because social cognitive deficits are central in schizophrenia and patients typically display impairments in social functioning.[25] The social aspect could therefore play a key role. Indeed, we found that the patients' ability to make inferences about others' mental states and their level of functioning were both specifically related to how they used information from others. This was also the case for the control subjects. Participants that were better at mentalizing and had a higher level of functioning relied more on cues about others' confidence to weight information from others. This was also the case when controlling for IQ. The results suggest that these patients and controls were more attuned to the cues from others and/or better at making inferences based on them and therefore more able to make use of them, and ultimately to function in the real world. Our results also suggest that it is important to keep in mind that social cognitive function presumably modulates the inference process in social contexts and may therefore shape hallucinations and delusions as well as social behaviour. Note, that although the basic inference mechanism may be the same, and our main study does not discriminate between a specific social or more general cognitive domain abnormality, our control experiment shows that placing the task in a social versus non-social context affects how the information is treated. In particular, we show that while directly experienced information is treated similarly in the two contexts, expressed confidence is treated differently according to whether it comes from an algorithm (non-social source) or a person (social source) and it is used more to inform decisions in the social context. In addition, the fact that neither context nor social cognitive function was related to the general tendency to overcount direct experience suggests that this specific abnormality, which may be central to schizophrenia symptomatology, is not closely related to how social information is processed.

The fact that patients focused more on directly experienced information as opposed to information from others compared to controls, could not be easily explained by general cognitive deficits. First, both patients and controls differentiated between high and low confident others. This is arguably the most complex part of the task and suggests that they did not have problems with processing the information provided. Second, both patients and controls often stated during debriefing that they put more trust in their own sample (higher certainty) than the others' decisions and not that they had problems with considering all the available information. Third, patients' working memory and executive function were not related to task performance. While participants' IQ was related to how they weighted others' confidence, overcounting of directly experienced information was not associated with a lower but a higher IQ in patients. It is unclear why this was the case, and we did not see this pattern in the control subjects.

Our findings are consistent with the idea previously put forward that patients are swayed by the evidence of their senses and tend to jump to conclusions based on this information, failing to take other—prior or in our case indirect—evidence into account.[3,14,28,52,53] This has been interpreted as a precision imbalance, where a reduction in precision at higher levels of the inferential hierarchy biases inference towards sensory evidence and away from prior beliefs,[27] or in the circular inference framework, that patients overweight and overcount sensory evidence and underweight prior beliefs.[3] This likely depends both on the context[6] and whether it concerns low-level sensory predictions or higher-order beliefs.[54,55] Importantly, it is an empirical question whether previous direct experiences are treated anything like indirect sources of information (e.g. social information). Another aspect is that patients do not always disregard social information. It presumably depends on whether they have access to directly experienced information or not,[56] as well as the level of processing.[57,58,59] Relatedly, patients may form strong beliefs under certain circumstances and are consequently reluctant to update them based on new information,[5,60] or as in our case, indirect information. Crucially, this new information typically comes from other people (disagreeing with the patient's view) and therefore will be under-counted and not challenge the fixed beliefs (delusions) that are rooted in direct experience. More generally, our findings point to the limitation of conceptualizing abnormal inference in schizophrenia as solely a problem of sequential belief updating of one type of information. In fact, a key abnormality may lie in how multiple sources of information are integrated simultaneously and in particular how directly experienced information is processed compared to other types of information, e.g. information from others. This only becomes evident when moving beyond one information source. Future studies could further investigate how different sources of information (direct experience versus indirect information) are treated during sequential belief updating.

Some limitations of this study should be mentioned. We cannot, based on the current data, determine whether circular inference is causally related to the formation of psychotic and negative symptoms. While it seems unlikely that overcounting of direct experience should stem from all types of schizophrenia psychopathology, it is possible that for instance putting less weight on information from others could be a consequence of the distrust experienced by patients with severe delusions. Future studies could assess circular inference in prodromal stages of the disorder or in high-risk groups as well as across diagnoses with or without psychotic symptoms. This could also circumvent the problem with antipsychotic medication being a potential confounder, although the evidence for this was weak. Similarly, investigations could be extended to the general population to get a better understanding of whether even low levels of overcounting of directly experienced information are related to subthreshold symptoms. Future studies could also investigate the purported underlying neural mechanisms of the circular inference model to provide further medium for hypothesis testing and establish a better understanding of the link between abnormal inference and the heterogeneous psychopathology of schizophrenia. In addition, although participants spoke of the agents as real people during debriefing, future studies could more directly assess to what extent the participants thought the agents were real and how this would impact task performance. Similarly, as different people may differ in their motivation to make accurate decisions,[61] future studies could directly compare how intrinsic versus extrinsic incentives[62] would affect task performance.

In conclusion, our findings provide further evidence of the benefit of using biologically informed computational models to critically investigate the underlying mechanisms of disorders like schizophrenia, beyond the immediate behavioural data. Our results suggest that hallucinations, delusions and negative symptoms could stem from the same underlying abnormality in inference, where directly experienced information is assigned an unreasonable weight and taken into account multiple times, so that even random and unreliable experiences may gain disproportionate significance. This may eventually lead to false perceptions (hallucinations), false beliefs (delusions) and deviant social behaviour (e.g. loss of interest in others, bizarre and inappropriate behaviour). This may be particularly problematic for patients with social cognitive deficits, as they to a larger degree may miss or fail to make use of corrective information from others, ultimately leading to worse social functioning.

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Competing interests

The authors report no competing interests.

Supplementary material

Supplementary material is available at *Brain* online.

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